Chapter

Perspective Chapter: Agroecology-Based Natural Farming in India

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Abstract

The current biospheric emergency, fueled by climate change and habitat loss, necessitates a re-evaluation of food production systems. This chapter advocates a crucial shift to natural farming, emphasizing crop diversity and interdependence. It proposes alternatives to the food production crisis, critiquing chemical-dependent conventional farming for its adverse impacts on land, yields, and sustainability. Natural farming, characterized by minimal inputs was presented as a sustainable method. Critical challenges in contemporary agriculture, including monoculture cropping and climate change, are examined. The chapter examines the evolution of natural farming in response to crises and government initiatives, delving into traditional practices, and indigenous knowledge, and exploring traditional food and seed systems for their nutritional value. Natural farming is showcased for its positive impact on soil biodiversity and its ability to counteract land degradation. The chapter highlights Andhra Pradesh's community-managed Natural farming for its role in generating public debates on food systems transformation. Acknowledging the urgent need for food system transformation, the chapter concludes with a call for research partnerships to guide natural farming's expansion, emphasizing collaborative efforts for sustainable advancement in India's agricultural practices.

Keywords: natural farming, seed pelletization, pre-monsoon dry sowing, 365 days green cover, traditional seeds, biostimulants, resilience, nutritional security

1. Introduction: Revisiting 'Food Systems' - India's learnings from natural farming

Agriculture is the key component of our food supply chain, and the methods employed to produce our food have far-reaching impacts on our health, the environment, and our economies. Agroecology involves the utilization of ecological principles and methodologies to sustainably improve and oversee soil fertility and agricultural productivity over the long term [1]. Several farming systems are partly or wholly aligned with agroecology. Some of these methods are reviewed from the angle of their potential to support food security in India. These techniques encompass: (i) Natural farming, also known as "Do Nothing" farming by Fukuoka; (ii) Biodynamic agriculture by Steiner, which has been implemented in India; (iii) Vermiculture developed by Appelhof and introduced in India; (iv) Natueco-culture by Dabholkar; (v) Zero Budget Natural Farming by Palekar; (vi) Rishi-Krishi by Deshpande; (vii) Agnihotra by followers of Gajanan Maharaj from Akkalkot; (viii) Panchagavya by Natarajan; (ix) Krishi-suktis and Vrikshayurveda from the wisdom of ancient and medieval Indian sages and scholars; (x) Compost tea introduced in India by Ingham; and (xi) Bokashi tea introduced in India by Higa [2]. Other ecologically harmonious agri-food systems are Organic Farming, Regenerative Agriculture, and Sustainable Farming.

The challenges of chemical-based current agri-food systems or conventional systems are well-documented. Organic farming and Agroecology-based Natural Farming are being tested and extensively in certain locations of India as alternatives to the current food system. Natural Farming is a holistic land management practice that leverages the power of photosynthesis in plants to close the carbon cycle and build soil health, crop resilience and nutrient density. Natural Farming constitutes a collection of agroecological techniques to rejuvenate soil vitality through emulation of natural processes. This comprehensive land management approach harnesses the potential of plant photosynthesis to complete the carbon cycle, enhancing soil quality, bolstering crop durability, and improving nutrient richness [3]. It places a greater emphasis on soil biology over soil chemistry, promoting practices such as multi-cropping and year-round soil cover and and incorporating a blend composed of cow dung and urine to stimulate the microorganisms within the soil ecosystem [4]. Natural farming practices are highly location-specific and vary greatly at different locations and even within a location, sometimes specific to practicing farmers. This chapter gives an account of perceptions of practicing farmers, promoters including governmental and nongovernmental agencies, formal scientists, and academicians in India on natural farming.

As a prelude, a brief comparative analysis of three distinct agri-food systems, namely, chemical-based conventional, organic, and natural farming systems, to various parameters is presented below.

Crop yields: Conventional Agriculture aims to maximize yields, often pushing the limits of land and resources, but it comes with inherent risk factors. Organic Farming strikes a balance, optimizing yields with a dependence on premium prices to compensate for yield gaps. Natural Farming, on the other hand, consistently demonstrates that maximization is possible without compromising sustainability.

Crop Production Inputs: In Conventional Agriculture, deficient plant nutrients are supplied through chemical fertilizers and the management of pests through chemical pesticides. This practice often leads to a loss of valuable soil carbon, humus, microbes, and earthworms due to plowing and other practices. Organic Farming relies on organic inputs for plant nutrition needs and biopesticides to avoid synthetic chemicals, promoting soil and environmental health. In Natural Farming, minimal external inputs are used. Inputs made on the farm by the farmers stimulate soil biology, avoiding harmful chemicals or bulky organic inputs.

Seed is also one of the main inputs that determine crop productivity. Conventional Agriculture frequently employs improved crop varieties and hybrids released through formal research systems including GMOs or transgenic crop seeds to enhance yield and quality of crop production. Organic Farming avoids hybrids and transgenic crop seeds that often need chemical inputs for claimed performance. Natural Farming also completely abstains from the use of the improved crop varieties and hybrids including GMOs or transgenic crop seeds. Traditional/Indigenous/Farmer varieties that are known to perform in specific geographies are encouraged to be used in natural farming.

Poly-cropping and Cropping Intensity: Conventional Agriculture typically practices minimal poly-cropping but encourages two or three crops (mostly mono-crops, sometimes mixed cropping with two or three crops) in a year for higher productivity. Organic Farming also follows similar cropping systems. In contrast, Natural Farming employs intensive polyculture (many crops ranging from 9 to 30), harnessing sunlight and soil moisture efficiently at various strata, which has the potential to significantly increase biodiversity, cropping intensity, and per unit land productivity.

Water Management: Conventional and Organic Agriculture primarily rely on rainfall or external irrigation sources often uneconomically for crop production. Natural Farming considers various sources of water, including water vapor in the atmosphere and "whapsa" water in the rhizosphere, enabling cultivation independent of rainfall with innovative practices of Pre-Monsoon Dry Sowing (PMDS) and 365-day green cover systems.

Net Income: Conventional Agriculture often results in negative net incomes, pushing farmers into crises. Organic Farming generally yields nominal to moderately positive incomes, depending on price premiums. Natural Farming shows highly positive economic impacts due to local, non-bulky inputs (Ghana and Drava Jeevamurt as against bulky Farmyard Manure encouraged in organic farming), high crop diversity, and reduced cultivation costs.

Consumer Health and Food Safety: Conventional Agriculture contributes to negative health impacts due to exposure to unsafe chemicals and the consumption of residue-laden foods. Organic Farming offers betterment of consumers' health by avoiding agrochemicals and providing residue-free foods. Similarly, Natural Farming prioritizes consumer health, ensuring residue-free, diverse, and nutrient-dense food options.

Impact on Natural Resources: Conventional Agriculture has a negative impact on natural resources, contributing to desertification, land degradation, water, and environmental pollution. Organic Farming demonstrates improvements by reducing soil, water, and air pollution and enhancing resource savings. Natural Farming stands out with a rapid and substantial improvement in natural resources such as soil health and agrobiodiversity, including conservation of the most important and dwindling resource, the water.

Demand, Consumption, Certification, and Marketing: Conventional Agriculture is commonly opted for, despite its shortcomings, in situations where no safer alternatives are available at a comparable price. Organic Farming caters to those who can afford premium pricing for certified safe foods. Natural Farming, unlike the others, lacks organized certification and relies on local trust, making it accessible to broader stakeholders, including economically disadvantaged communities.

Each food system offers unique advantages and challenges. Conventional Agriculture seeks maximum yields but faces risks and sustainability concerns. Organic Farming balances yield optimization with sustainability and safety. Natural Farming, with its minimal external inputs, showcases high productivity, sustainability, and affordability while prioritizing local trust over certification. The choice among these systems depends on various factors, including environmental goals, economic considerations, and consumer preferences.

2. India's agriculture research and development

The Department of Agriculture and Farmers Welfare, which operates within the framework of the Ministry of Agriculture, serves as the central entity entrusted with the advancement of India's agricultural sector [5]. Behind the growth lies a history of India's efforts in increasing production post-Green Revolution. Agriculture holds a significant position in India's economic landscape, currently ranking as one of the world's top two agricultural producers [6]. The agricultural sector in India contributes to 18% of the nation's gross domestic product (GDP) and engages nearly 54.6% of the nation's labor force [5]. As stated by the economic data of the financial year 2018–2019, agriculture has acquired 18% of India's GDP. The agriculture sector of India occupies almost 43% of India's geographical area.

The growth achieved in Indian agriculture is driven by postindependence conditions such as food insecurity, lack of rural employment, hunger, and malnutrition. On the eve of the first plan (1951–1956), agriculture was hopeless and deplorable. Farmers were in heavy debt to the village moneylenders. They had small and scattered holdings. They had neither the money nor the knowledge to use proper equipment, good seeds, and chemical manures. Except in certain areas, they were dependent upon rainfall and upon the vagaries of the monsoons. Productivity of land as well as of labor had been declining and was generally the lowest in the world. Seventy percent of the active workforce was involved in farming, and although the nation did not produce enough food grains to meet its own needs, it had grown reliant on importing food grains [7].

Efforts were specifically directed toward enhancing food and cash crop supplies. The Grow More Food Campaign in the 1940s and the Integrated Production Programme in the 1950s targeted these aspects separately. India's 5-Year Plans [8], focused on agricultural development, quickly followed suit. Under government oversight, activities such as land reclamation, land development, mechanization, electrification, and chemical usage, particularly fertilizers, were implemented. An integrated approach to agriculture, emphasizing comprehensive actions instead of isolated aspects, was developed [8]. These government policies catapulted India into one of the world's leading producers of wheat, edible oil, potatoes, spices, rubber, tea, fish, fruits, and vegetables. To support agricultural research, various institutions were established under the Indian Council of Agricultural Research. Additionally, organizations like the National Dairy Development Board (established in 1965) and the National Bank for Agriculture and Rural Development (established in 1982) facilitated cooperative formation and improved financing [8].

The Green Revolution's focus was on increasing the yields, increasing the acreage, and improving overall production, thereby addressing food security. In India, the initiation of Green Revolution was orchestrated under the leadership of geneticist Dr. M. S. Swaminathan [9]. It started around the 1960s and helped in increasing food production in the country. The Indian agriculture scientists from ICAR and other institutions were sent to advanced universities in the USA regarding new agricultural technologies and their application with the support of USAID and Public Law 480 (PL 480). ICAR institutions undertake aid and promote and coordinate education, research and its application in agriculture, animal science, fisheries, agroforestry, and allied sciences.

The whole ecosystem of India's current agriculture research, extension, and partnerships (public–private) is guided and rooted in green revolution. ICAR launched the All India Coordinated Research Project -National Seeds Project (Crops) in 1979–1980 aimed at the production of basic seed with a separate project-coordinating unit dedicated exclusively to nucleus and breeder seed production and to carry out seed technological research. This was elevated to the directorate paving the way for the establishment of the Directorate of Seed Research (DSR) in 2004. Around 4500 seed varieties are developed; their multiplication is restricted to a few varieties. Krishi

Vigyan Kendras (KVKs) are the centers for agriculture extensions created by ICAR to serve as knowledge resource centers at the district level. They are an integral part of the National Agricultural Research System (NARS) and serve as the link between the NARS and the farmers. KVK's whole work such as technology assessment and demonstration for its application, capacity development, and crop advisory services is tuned to chemical farming. Farmer participation in the development and dissemination of technology has emerged as an important theme in extension practice over the years. The synthetic inputs, especially fertilizers and pesticides, are produced and distributed by private entities.

3. Crop genetic exploitation

Crop yield, quality, and pest resistance were mainly achieved through the exploitation of genetic resources and breeding approaches during the Green Revolution period. To address the intricate challenges confronting our planet, including rapidly changing climate patterns, food and nutritional insecurities, and the ever-increasing world population, modern agricultural practices turned to the development of hybrid varieties of crops. The establishment of a commercially viable system for producing hybrid seeds has profoundly impacted our contemporary scientific comprehension of crops and the agricultural industry. Hybrid crops brought significant advantages for bolstering the economic benefits primarily owing to the phenomenon of heterosis, which confers superiority over diverse parent varieties. Leveraging genetic mechanisms to harness heterosis has facilitated enhanced productivity, quality improvement (depending on the specific goals), and reduced seed production costs. It effectively achieved the goal of developing crops with higher yields and improved quality across the crops. Thus, crop yields and farmers' incomes enhanced, ensuring food security.

Improved varieties and hybrids developed still had certain limitations of not having sources of resistance to pests and diseases in the germplasm or genetic stocks of such crops. Hence, scientists explored the possibility of using the resistant sources outside the plant genome. The first success story in India was on incorporating *Bacillus thuringiensis* gene (Bt gene) into cotton varieties and hybrids. Such varieties having transgressed genes from bacteria to plant are called transgenic varieties or Genetically Modified Crops (GMCs). Thus, these cotton transgenic hybrids dominated by the private industry suppliers not only enhanced the cotton yields but also initially needed very less pesticides for cotton cultivation. Currently, over 90% of cotton cultivated area is occupied by Bt hybrids. Even this approach had limitations that the same hybrid cannot resist different bollworms and races within the same pest, and thus, continuous development through gene stacking became necessary.

Seed systems in India focused largely on improved varieties, hybrids, and transgenic crops. Public institutions such as research institutions and state agricultural universities play a key role in the seed supply chain by providing foundation and breeder seeds of the improved varieties and parental lines of hybrids. A major chunk of the seed supply of these varieties is in the hands of private industry. Traditional varieties are largely ignored by the formal seed systems. A small section of NGOs, farmer networks, and a few state departments of agriculture are only involved in traditional varieties of seed supply.

The modernization of agriculture meant that newer seed varieties were continuously developed by public and private sectors independently and through partnerships. The establishment of organizations such as The International Union for the Protection of New Varieties of Plants (UPOV) in 1961 has introduced a regulatory environment in the name of protecting plant varieties and encouraging the development of new varieties of plants. In 2001, India became the first country worldwide to introduce the Protection of Plant Varieties and Farmers' Rights Act. Additionally, India established an implementing authority to safeguard the rights of both farmers and breeders [10]. NITI Aayog (National Institution for Transforming India) in its working paper reviewed the status of agriculture in India and observed that "Despite the noteworthy increase in per capita food production, some sections of the population still suffer from undernutrition and malnutrition. There is a significant change on the demand side, with consumer preferences shifting towards healthy, safe, trait-based and quality food and bios. These changes indicate that the future of agriculture (and those engaged within) will face profound transformation in the coming decades" [11]. Authors believe that not only the exploitation of frontiers sciences but also the promotion of natural farming will address the issues of undernutrition, malnutrition, and access to healthy safe and quality food. Further natural farming will improve the environment and soil quality through sustainable agricultural practices.

4. Challenges in the current approaches to agriculture

Monoculture farming refers to cultivating a single crop on a large percentage or all of the farm, season after season to maximize production and profitability by growing a single crop. Mono-cropping practices contribute to the depletion of biodiversity; cultivation practices such as repeated and deep plowings lead to soil compaction; chemical application reduces or kills the microbial diversity that contributes to the decline in the availability of plant nutrients in the soil. Plowing, grazing, excess irrigation, keeping lands fallow, overgrazing, wind and water erosion, burning crop residues, and, finally, the use of excessive chemical fertilizers have almost destroyed the biological processes that stabilize soil structure and maintain its fertility, temperature, and water-holding capacity.

In India, soil erosion leads to the deposition of silt in reservoirs, resulting in a gradual annual reduction in reservoir capacity estimated at 1 to 2 percent. This, in turn, adversely affects irrigation in the surrounding areas. The National Academy of Agricultural Sciences (NAAS) estimates that water erosion causes an annual production loss of 13.4 million tons for major rainfed crops in India, translating to a financial loss of Rs. 205.32 billion [4]. Another area of huge concern is flood irrigation practices promoted by conventional farming. Starting from the 1970s, there has been a substantial rise in the extent of land using groundwater for irrigation. India stands as the globe's leading consumer of groundwater, with an annual withdrawal of 250 cubic kilometers [12]. Approximately 20 million irrigation wells are scattered across India [13]. India is among the top 10 pesticide-consuming countries in the world.

Climate change also induced extreme weather events such as floods, forest fires, and heatwaves to name a few that have destroyed crops, reduced harvests, and impoverished farmers. The anticipated temperature rise of 1–2.5 degree Celsius by 2030 is expected to have profound consequences on crop yields. Increased temperatures may result in a decrease in crop duration, enable modifications in photosynthesis, raise crop respiration rates, and impact pest populations [4]. Farmers and laborers are at risk of occupational hazards that are being exacerbated by climate change. A combination of the above factors has significantly reduced the quality of food produced through chemical farming. Due to intense, mismanaged farming, soil nutrients are declining. Nitrogen stores have decreased by 42%, phosphorus by 27%, and sulfur by 33%.

According to the National Academy of Agricultural Sciences (NAAS) estimates, India experiences an annual soil loss rate of approximately 15.35 tons per hectare, leading to a depletion of nutrients ranging from 5.37 to 8.4 million tons [4]. The decline of biodiversity is adversely affecting the functioning and stability of ecosystems [14] and diminishing human well-being by decreasing the services that ecosystems can provide for people [15]. To quote an example, the decline of the honeybee, a major pollinator of many crops, is impacting agriculture. As per FAO (Food and Agriculture Organization), bees are responsible for pollinating 71 out of the 100 crops that constitute 90% of human food, with an estimated annual economic value of up to $200 billion [16].

5. Current agriculture crisis

The green revolution that was initiated during the 1960s helped to cross the edge of famines and ensure food security in most developing countries including India. Presently, India not only fulfills its own food requirements but also maintains substantial food grain buffer reserves and holds the distinction of being the world's largest exporter of rice [17]. Seventy years ago, the agricultural strategy centered on expanding land under cultivation, clearing additional areas for farming, and harnessing the advantages of science and technology to boost production. This has led to increased use of synthetic fertilizers and agrochemicals, modified seed varieties that produce higher yields, and reengineering of agronomy practices that manipulate soil and water systems. It has contributed to the degradation of land, soil health, and well-being of people due to the prolonged use of synthetic fertilizers and chemicals.

The extreme and indiscriminate adoption of modern agricultural technologies for higher crop yields has shown its baneful effect over the years. The consequence is deterioration of soil fertility; currently, 75% of the earth's land is degraded, losing 36 billion tons of soil annually. The Food and Agricultural Organization (FAO) indicated that due to soil degradation, only 60 harvesting years are left, which is a major concern for the growing population [18]. There is a significant reduction in soil organic carbon to 0.3 percent, and its magnitude of loss may be from 10 to 50 tons C/ha [19, 20]. The soil with low soil organic carbon yields results in low agronomic yield.

Fertilizer use is highly skewed among the states in India [21], and fertilizer consumption has increased from 12.4 kg/ha in 1969 to 133.4 kg/ha in 2020. The imbalanced use resulted in a declining fertilizer crop response ratio from 123.47 kg grain/kg in 1970 to 3.70 kg grain/kg in 2005. During 2020–2021, the Government of India spent Rs. 127,921 crore on fertilizer consumption. India has the biggest pesticide industry in Asia with consumption worth Rs. 6000 crores. The extreme consumption of agrochemicals has led to water, soil, and air contamination, thereby reducing crop productivity [22].

Conventional chemical farming increased the cost of cultivation and led Indian farmers into debt trap [23]. This chemical farming method uses fossil fuel and water at an unsustainable rate, and agriculture in India consumes 90% of the water of which 80% is used in the production of rice, wheat, and sugarcane [24, 25].

Climate change can affect agriculture in a variety of ways. Climate change induced yield loss between 4.5 to 9% leading to a loss of 1.5% of GDP on an annual basis. Climate change is a marginal negative gain over the years, and this is affecting crop production and livestock management [26]. Climate change is going to have an adverse effect on major field crops like rice, wheat, soybean, millets, and sorghum, whereas the impact on fruits and vegetables is widely varied, as it depends on the latitude and region of cultivation. In the case of the livestock sector, there is going to be a reduction in pasture productivity along with decreased growth rates of productivity [27] (Bhattacharyya et al., 2020). Greenhouse gas (GHG) emissions stemming from the food system globally are high, notably attributed to energy consumption throughout the food supply chain [28]. In India, 14% of the total GHG emission is contributed by the agricultural sector [29]. Climate change has a significantly greater risk to the natural and human systems with 2°C global warming above pre-industrial temperatures compared to 1.5°C; biodiversity can lose twice as high under this condition [30]. Elevated temperatures can disrupt plants' capacity to obtain and utilize moisture efficiently [31]. Global warming is likely to increase rainfall; the net impact of higher temperatures on water availability is a race between higher evapotranspiration and higher precipitation. Globally, the overall impact of baseline global warming by the 2080s is a reduction in agricultural productivity (output per hectare) by 16%.

There are numerous drawbacks to the use of industrial chemicals that are not organic, especially in terms of their environmental footprint. These harmful synthetics used in massive amounts contaminate plants grown for human consumption. They also result in a vast heap of residue in the soil postharvest, which not only permeates the soil but also degrades groundwater quality. They lead to excessive fertilizer use, habitat destruction, ecological degradation, soil fertility loss, carbon emission, and so forth. Land-use change (LUC) resulting from the expansion of agriculture constitutes a significant origin of anthropogenic GHG emissions [32]. The combined contributions of agriculture, forestry, and other land use (AFOLU) accounted for approximately 23% of global anthropogenic GHG emissions during the period of 2007–2016 [33]. Moreover, the variability of emissions from deforestation and from cultivated organic soils drive, on average, 42% of the variance in product agricultural GHG emissions, according to the meta-analysis conducted by Poore and Nemecek (2018). The existing conventional agri-food systems that rely heavily on chemical inputs, along with other industrial activities, have had a detrimental impact on various aspects of our ecosystem. These include causing distress to farmers, degrading the quality of food, soil, and water, and contributing to the worsening of the overall health of humans, animals, soil, and the planet as a whole.

6. Natural farming in India

Natural farming origins can be traced to when Mokichi Okada proposed the concept of "nature farming" in 1935 [34]. While Masanobu Fukuoka popularized the term shizen noho (meaning natural farming in English), Okada was the first to introduce farming without fertilizers and pesticides. Natural Farming adopts a low-input, climate-resilient farming approach that promotes the utilization of inexpensive, locally sourced materials while eliminating the need for artificial fertilizers through cost-effective cultivation methods [35]. Natural Farming helps revive soil organic matter and biomass regeneration and helps in ecosystem services.

In India, the farm-based livelihood interventions under DAY-National Rural Livelihood Mission (DAY-NRLM) were started first under Mahila Kisan Shasaktikaran Pariyojana (MKSP), launched in 2010–2011 as a sub-component of DAY-NRLM. MKSP has made a significant change in the use of new scientific knowledge and practices intended to revive age-old traditional knowledge, recycle biomass, restore soil fertility, improve seed and sovereignty, protect plants, and improve storage practices and technology intervention that are easy to adopt and scale up. Since 2014–2015, India has had a National Mission for Sustainable Agriculture (NMSA) to promote sustainable agriculture. It consists of several programs focusing on agroforestry, rainfed areas, water and soil health management, climate impacts, and adaptation. However, the allocation for the National Mission for Sustainable Agriculture (NMSA) within the Ministry of Agriculture and Farmers Welfare (MoAFW) budget is minimal, accounting for just 0.8%.

Natural farming has been gaining popularity in India as a sustainable farming methodology since 2015. Approximately 3.8 million hectares, equivalent to 2.7% of the total agricultural land in India, is dedicated to organic or natural farming methods [35]. The government is pushing natural farming across India, with schemes like Bhartiya Prakritik Krishi Paddati (BPKP) and Zero-Budget Natural Farming (ZBNF). The BPKP targets to cover 12 lakh ha in 600 major blocks of 2000 ha in different states over a period of 6 years. Mission LiFE was officially launched by the Hon'ble Prime Minister of India on October 20, 2022. It envisions an India-led global grassroots movement aimed at inspiring both individual and collective actions for the protection and preservation of the environment [36].Mission LiFE has a comprehensive and non-exhaustive list of 75 individual LiFE actions across seven categories, and "Natural Farming" is one among them.

Starting in 2016, the Government of Andhra Pradesh (GoAP) has been executing the AP Community-managed Natural Farming (APCNF) program through Rythu Sadhikara Samstha (RySS), a government-owned Section 8 company. APCNF adopts an approach that is farming in harmony with nature and works with more than a million farm and farmworker families. By harnessing the social capital that exists in the form of women's self-help groups and their federations, APCNF promotes farmer-to-farmer experiential knowledge dissemination [37].

However, some of the key challenges identified in the adoption of Natural Farming are farmers' mindsets who are used to chemical farming for over 60 years and require hand-holding support from a credible extension system as the current extension follows a top-down approach. The state agriculture departments, the agriculture universities, and the agriculture Research system and banking are all geared toward chemical agriculture, a real hurdle in weaning farmers away from chemical farming. An account of the key approaches, science-based perceptions, traditional knowledge, and innovations that contribute to natural farming promotion in India is provided below.

7. Evolution of traditional food Systems in India

India's agriculture has been a rich tapestry of diverse and harmonious farm practices since time immemorial. Food systems are best understood as a way of life and an experience, as they are deeply embedded in and shaped by the landscape, ecology, culture, social exchange, economy, politics, and people's lived experiences. Agricultural systems are a part of food systems as recognized even in mainstream definitions of food systems: "This system encompasses all aspects, including the environment, individuals, resources, processes, infrastructure, institutions, markets, trade, and the activities associated with the production, processing, distribution, marketing, preparation, and consumption of food, as well as the resulting socio-economic and environmental consequences," High-level Task Force of Global Food and Nutrition Security, October 2015 [38].

A country with a population of 1.2 billion people, India is home to 705 indigenous communities [39]. Each of these communities has its unique food system that includes a wide variety of indigenous foods, contributing to increased dietary diversity [40, 41]. Historical records (9000–2000 BCE) from India indicate the early cultivation of wheat, barley, horse, sheep, goat, elephant, and cattle. These practices encompassed tasks such as threshing, row-based crop cultivation—either in pairs or in groups of six—and grain storage in granaries. The crops under cultivation ranged from cotton, hemp, and sugarcane, eventually transitioning to rice. The Indus Valley Civilization marked the advancement of irrigation and water storage techniques, including the creation of reservoirs that facilitated mixed farming and plowing using animal power.

Turning to the realm of agriculture, the prevalent method in earlier Indian practices, particularly within tropical regions, was "shifting agriculture" system, recognized under various local names such as "swidden," "slash and burn agriculture," rotational bush fallow agriculture, and commonly known as "jhum." This comprehensive agricultural approach involves alternating short-crop phases, typically lasting one or 2 years, with periods of natural or minimally altered vegetational fallow. Rather than pursuing short-term maximization, the emphasis is on long-term yield management. Shifting agriculture methods uphold diversity in cropping phases through mixed cultivation. Perennial shrubs and trees are staggered in time and restricted to the fallow phases within the forest cycle. In this framework, agricultural ecosystem functions such as nutrient cycling and pest dynamics are governed through the intricate interplay of cropping and fallow stages. The system's stability hinges on maintaining a minimum agricultural cycle length—defining the duration of the fallow period before the return to the phase fostering soil fertility recovery, a factor directly impacting economic yield. These strategies safeguard food's nutritional density, ensuring a wealth of minerals and vitamins. However, the abundant diversity inherent in traditional agroecosystem types, present until more recent times and still existing in numerous remote Indian locales, faces rapid depletion. Furthermore, the profound traditional ecological knowledge and technologies nurtured by these indigenous societies are fading due to the dominance of mainstream monoculture practices propagated by conventional farming.

8. Nutritional value of indigenous food systems (IFs)

India boasts a rich diversity of native foods (IFs) that hold deep cultural significance within local and ethnic communities. Recently, the FAO has emphasized the establishment of a "Global Hub on Indigenous Food Systems," renewing global interest in IFs due to their potential to enhance food security and promote biodiversity on a global scale. These foods are esteemed for their substantial nutritional value, offering significant prospects for improving health and nutrition [42].

Indigenous communities gather and cultivate local IFs from a wide range of food sources, including edible greens (such as leaves, stems, and shoots, including marine algae), root vegetables (encompassing true roots and underground storage organs like bulbs, corms, tubers, and rhizomes), fleshy fruits (such as berries, pomes, and drupes), wild mushrooms, grains, seeds, nuts, as well as harvested fish, insects, and game [43]. In traditional practices, farmers not only conserved their own seeds but also engaged in a system of barter or exchange, sharing their seed stocks with one another.

An analysis of several articles offering scientific evidence, utilizing selective indicators, has furnished insights into the nutritional worth of indigenous foods. Documentation has been made regarding the nutritional values of 508 IFs consumed across 15 states and 1 UT of India. Jharkhand leads at 28%, followed by Meghalaya at 15% and Kerala at 11%. These IFs fall into distinct categories: (a) 63 cereals, (b) 36 nuts and legumes, (c) 154 green leafy vegetables (GLVs), (d) 98 other vegetables (including mushrooms), (e) 51 roots and tubers, (f) 66 fruits, and (g) 40 flesh foods (including livestock and wild game). Details and units were extracted from each study for all macronutrients, micronutrients, and anti-nutrients, meticulously reviewed and systematically organized according to food groups.

The extracted data created a compendium detailing the nutrient (n = 508) and antinutrient (n = 123) content of IFs, followed by calculating antinutrient-to-mineral molar ratios for 98 IFs to predict mineral bioavailability. Green leafy vegetables (n = 154) boasted the highest nutritive values, trailed by other vegetables (n = 98), fruits (n = 66), cereals (n = 63), roots & tubers (n = 51), and nuts and legumes (n = 36). Many IFs surpassed conventional foods in nutritional content and were rich (i.e., >20% of Indian recommended dietary allowances per reference food) in iron (54%), calcium (35%), protein (30%), vitamin C (27%), vitamin A (18%), zinc (14%), and folate (13%).

Within cereals, 49 local rice landraces' nutritional values have been documented, cultivated, and consumed by various indigenous communities. Varieties of red rice (Kba-baswoit, Ladhan, Kba-bakut, Kba-Iwai, and Kba-stem) and sticky rice (Kba-shulia) in northeast India exhibit high iron (3.1–6.1 mg/100 g) and folate content (65–1203 μg/100 g). Several millet varieties also feature elevated iron (3.9–19.3 mg/100 g) and calcium (264–364 mg/100 g) content. Indigenous legumes from Jharkhand, Kerala, and Andhra Pradesh serve as substantial sources of calcium (260–945.4 mg/100 g), iron (3.6–38.6 mg/100 g), and zinc (3.4–7.7 mg/100 g), respectively. Among indigenous nuts, Khasi tribe of Meghalaya consumes varieties with remarkably high calcium levels (1020–1540 mg/100 g), while Perilla (Perilla frutescens), an oilseed from Meghalaya and Manipur, boasts elevated iron (8.3–9 mg/100 g) and zinc (4.7–5.02 mg/100 g) levels.

9. Bringing traditional methods back into farming

In the current Indian food landscape, inadequate and suboptimal sustenance plagues around 14% of the roughly 900 million rural and 460 million urban residents. Reincorporating traditional methods, such as millet cultivation, aims to enhance diet diversity. Beyond health advantages, millets promote environmental health due to their minimal water and input needs. The United Nations has declared 2023 as the International Year of Millets [44], a bid backed by the Indian Government to boost millet production and consumption. To enrich food's nutritional value, the Indian government is reviving traditional seed variations. Natural Farming methods help decrease exposure to harmful chemicals, subsequently lowering the incidence of related illnesses and the associated healthcare expenses. Furthermore, the AP

Community-managed Natural Farming (APCNF) initiative works in partnership with villages to encourage a varied diet and guarantee access to public health and nutritional services [45]. Several studies are underway to examine both crop quality when Natural Farming principles are applied and the health outcomes of APCNF.

10. Traditional knowledge recognition and promotion

Traditional societies since centuries have accumulated a whole lot of empirical knowledge from their experiences while dealing with nature and natural resources. Traditional wisdom is rooted in the fundamental understanding that humans and nature are interconnected and constitute an inseparable unity [46]. This is widely reflected in their attitude toward plants, animals, rivers, and the earth. Traditional Ecological Knowledge (TEK) encompasses the knowledge, innovations, and practices of indigenous and local communities across the globe. TEK is passed down orally from one generation to the next and is regarded as the intangible heritage and intellectual property of these communities [47]. It tends to be jointly closely held and sometimes takes the shape of stories, songs, folklore, proverbs, cultural values, beliefs, rituals, community laws, native languages, and practices. Of the 3800 culturally distinct communities living in India, over 50 million tribals have significantly contributed to the traditional ecological knowledge. Over 9500 wild plants are recognized to be of ethnobotanical value, of which over 7500 are used as medicines, 3900 as edibles, 500 for fiber, 400 as fodder, and 300 as pesticides.

The concept of villas as an ecosystem among traditional societies involving agriculture, animal husbandry, and the domestic sector including forest and forest-related activities such as hunting and gathering food and fodder, fuel, and medicinal plant collection and traditional farming practices such as shifting agriculture forms the basis for "natural farming." The manner in which traditional societies observe and interact with the biodiversity in their surroundings, considering both spatial arrangement and temporal dynamics, holds substantial importance in maintaining ecosystem stability and resilience, particularly in the context of land use management [46]. Traditional forest management hinges on the active involvement of local communities, with social justice and equity serving as fundamental principles in the management of bioresources [48]. Most indigenous natural resource management system needs little or no external input, is flexible, and evolves with time for which an in-built mechanism in the form of traditional institutions is in place.

The Apatanis of Arunachal Pradesh practice an ecologically efficient form of wet rice cultivation and have evolved practices by which varieties are determined according to nutrient status at a given location. The whole village system is largely dependent upon the recycling of village waste resources. Closer to the village, they emphasize a pure crop of high-nutrient uptake/low use of efficient rice cultivation. Combining mixed cropping strategies in space and time with traditional weed management strategies shifting agriculture farmer of northeast India ensures effective check on nutrient loss during the cropping phase. They practice weed management rather than weed control. Even the pulled-out weeds are put back into the system. They developed sophisticated wet rice cultivation systems tailored to variations in soil fertility and water resources, aligning them with traditional management practices suitable for the specific landscape [46].

A study conducted by Tiwari et al. [49] in Meghalaya underscores the role of traditional ecological knowledge held by tribal communities in promoting environmental sustainability. This knowledge has naturally evolved through their harmonious coexistence with their surroundings. The practice of allocating protected forest areas, such as "sacred groves, village restricted forests, village supply forests, clan forests, and other traditionally managed forests," which collectively constitute approximately 90% of Meghalaya's total forested land, empowers tribal communities to nurture and safeguard forests and trees in proximity to their settlements, near water sources, on steep slopes, and other ecologically sensitive regions [49].The cultivation of home gardens for example by traditional societies demonstrates structural variations, functional differences and spatial complexity, depending upon the social and ecological settings. Emulating a natural forest, these gardens consist of densely arranged economically significant trees, shrubs, and herbs in compact plots ranging from 0.5 to 2 hectares, often encompassing more than a hundred different species [46].

There is growing evidence of the role of traditional knowledge in responding to climate change. The IPCC's 4th Assessment emphasized the significance of indigenous knowledge and crop varieties in the context of adaptation [50]. UNU-IAS has recently recognized over 400 instances of indigenous peoples actively participating in climate change monitoring, adaptation, and mitigation. These cases encompass a wide array of effective strategies [51]. Studies conducted by IIED and its collaborators, involving indigenous communities in Peru, Panama, China, India, and Kenya [52], have revealed a strong and intricate connection between traditional knowledge and traditional crop varieties, highlighting their mutual interdependence [53]. The maintenance and transmission of TK depends on the use of diverse biological resources (wild and domesticated), and the reintroduction of traditional varieties can revive related traditional knowledge and practices. In a significant development, the 15th Conference of Parties (COP15) to the United Nations Framework Convention on Climate Change (UNFCCC) in 2010 endorsed a decision concerning "Enhanced action on adaptation" [53]. This decision underscored the necessity of incorporating "traditional and indigenous knowledge" alongside the best available scientific knowledge in addressing climate change challenges.

Natural Farming that is being practiced and adopted has drawn 5 important lessons from traditional ecological knowledge. First, Natural Farming principles are built on knowledge about resilience against drought and pest resistance. Traditional farmers are able to identify resilient crop species and varieties for adaptation. Traditional farmers are also plant breeders actively experimenting on farms. Thirdly, farmers did seed selection for preferred and adaptive characteristics, and fourth, knowledge about wild crop relatives is borrowed from times immemorial.

In Kenya, particularly among the Mijikenda community, certain wild food plants gain popularity among farmers when conventional crops face failure [53]. The fifth and perhaps most crucial aspect is that traditional agricultural practices play a vital role in preserving essential resources necessary for resilience and adaptation, including biodiversity, water, soil, and nutrients. Traditional knowledge can assist in predicting local weather patterns, anticipating extreme events, and offering accessible information to farmers, often at a scale more pertinent and practical at the local level than advanced modeling approaches [53].

11. Natural farming in India

Natural Farming offers a viable alternative to current conventional farming practiced in India; however, it requires systemic changes in both institutional—government and private—sectors. This type of farming practices based on ecological principles and laws of nature is being advocated by the United Nation to promote sustainable food systems to enhance food and nutrition security. Natural farming is being promoted through Bhartiya Parmparagat Krishi Pariyojna (BPKP) in India to enhance production, sustainability, saving of water use, and improvement in soil health and farmland ecosystem. The government has established the National Mission on Natural Farming (NMNF) as a distinct and autonomous scheme starting from 2023 to 2024. This expansion builds upon the Bhartiya Prakritik Krishi Paddati (BPKP) and is allocated a budget of Rs. 459 Crores [54].

The National Mission on Natural Farming (NMNF) is set to encompass an area of 7.5 lakh hectares through the establishment of 15,000 clusters. Farmers who express their intention to practice natural farming in their fields will be enrolled as members within these clusters. Each cluster will consist of 50 or more farmers, collectively managing 50 hectares of land or more [55]. Furthermore, each cluster can encompass a single village or span across 2–3 adjacent villages within the same gram panchayat. The central government aims to establish 15,000 Bhartiya Prakritik Kheti Bio-inputs Resource Centres (BRCs) to facilitate convenient access to bio-resources, with a focus on cow dung, urine, neem, and culture as essential components. These bio-input resource centers will be established alongside the proposed 15,000 model clusters for natural farming [55]. The Ministry of Agriculture is actively engaged in the extensive training of master trainers, “champion” farmers, and practicing farmers in natural farming methods. This training is facilitated through the National Institute of Agricultural Extension Management (MANAGE) and the National Centre of Organic and Natural Farming (NCONF) [56]. The Andhra Pradesh Community-managed Natural Farming (APCNF) program, initiated by the Rythu Sadhikara Samstha (RySS) under the Department of Agriculture (Natural Farming wing), Government of Andhra Pradesh, since 2016, has yielded valuable insights into extension and training, social capital, and research. APCNF model advocates for best-practicing cadres as champion farmers in taking the lead farmers and other farmers as apprentices on the journey of NF, till they take charge of their journey (over 2–3 years). An internal community resource person (1 per 100 farmers) leads the transformation in 1 or 2 clusters, each cluster covering 3–5 Gram panchayats. APCNF co-opted educated young practitioners as input shop owners, farmer entrepreneurs, and local individual and collective enterprises in extension. Natural Farming fellow (agriculture graduate with NF practice. 1 per 1–2 blocks) also extends support in transformation. A per capita cost of Rs 15,000 needs to be spent on each farmer for over 5–7 years.

To scale Natural farming based on farmers’ traditional knowledge and their innovative approaches, it requires committed funds from the Government of India. This requires mainstreaming Natural Farming in the Agriculture ecosystem of India and engaging the Indian Council for Agriculture research, the private sector, financial institutions, and others to develop digital architecture. APNCF is advocating for bringing changes in curriculum in schools and universities to introduce Natural Farming to make NF knowledge, skills, and tools available for behavioral change among farmers. Realizing the need for dedicated effort in NF research, evidence, knowledge, and learning domain, the Government of AP and RySS have set up the Indo-German Global Academy for Agroecology Research and Learning (IGGAARL, Academy) in Pulivendula. The academy has been set up by the State Government of AP, with the support of a 20 Million Euros Grant from Government of Germany. The academy aims to offer certified learning to 1,00,000 NF practitioners, 10,000 farmer scientists through NF degrees, and 1000 interns (graduates and post graduates in agriculture, microbiology, management, social work, etc.). The flagship program of the academy learning is the four-year Farmer Scientist Course. Seventy-five percent of the learning in FSC is through practical and experiential learning through their own field, own experiments, training peer farmers, and transforming their village into climate-resilient villages. This is facilitated by conceptual inputs in the classroom and supported by intense mentoring. Overall, the Indo-German Global Academy for Agroecology Research and Learning strives to transform farming practices, generate farmer-led evidence, and establish a global knowledge platform on natural farming.

The AP Community-managed Natural Farming (APCNF) has achieved recognition as a National Resource Organization (NRO) by the National Rural Livelihoods Mission (NRLM), operating under the Ministry of Rural Development, Government of India. This recognition positions APCNF to assist other State Rural Livelihoods Missions (SRLM) in their efforts to promote Natural Farming transformations. APCNF has entered MoUs with Madhya Pradesh, Meghalaya, and Rajasthan SRLMs and is providing technical support to 12 more states.

12. Climate change adaptation and mitigation

Climate change can have diverse impacts on agriculture. As per NITI Aayog, India confronts severe water stress, with 600 million people experiencing high to extreme levels of stress and almost 70% of the water sources being contaminated. Additionally, the Indian Space Research Organization reports that over 30% of the land is undergoing land degradation and desertification. Furthermore, average land productivity currently stands at one-fourth of its potential capacity [57]. Several studies anticipate significant reductions in global crop yields, ranging from 20% to as much as 90%, within a relatively short span of a few decades due to the combined effects of climate change, soil erosion, and the decline in critical pollinator populations [58].

APCNF has introduced pre-monsoon dry sowing, a practice to be followed for covering the soil with 20 diversified crops during pre-monsoonal season to rebuild the soil health. PMDS cropping system improves the soil's physical, chemical, and biological properties. PMDS has not only improved crop diversity but also significantly contributed to graduating farmers to adopt 365 days of green vegetation. Farmers developed unique A-grade models that followed all important principles of natural farming—365 days of green cover, diverse live plants at all times, relay cropping, higher land equivalent ratios, minimizing tillage, crop residue mulching, pelleting seeds, use of biostimulants to activate soil biology, use of indigenous seeds, and so on. All these practices implemented in the same plot of land create an excellent model of climate-resilient farming. Around 300 model farmers are being trained to take this concept to a large number of farmers. These innovations have worked in dryland regions of Ananthapuramu district.

Natural Farming has a scientific grounding in agroecological knowledge and practice. It posits that the powers of nature can be harnessed in ways that enable humans to extract agricultural products from soil systems in a sustainable way while maintaining and improving the ecosystem services of natural resources. This requires supporting regenerative processes within soil systems that replenish the stock of soil organic carbon and sustain below- and aboveground processes that compensate for the export of nutrients in food production. Regenerative agriculture is based on management practices that leverage plants' power of photosynthesis to restore the carbon cycle and water cycle and to build up soil health, crop resilience, and nutrient density in food. In natural farming, the existing processes are enhanced by cutting-edge research on soil biology and ecology. This hypothesis clarifies the functions of plants' microbiomes, the importance of root exudates, and the potentials and limits of biostimulants and bioinoculants. It also sheds light on the respective roles of bacteria, fungi, and viruses in plant health.

13. Customization of pre-monsoon dry sowing (PMDS) in APCNF

In 2018, APCNF leveraged poly cropping to pioneer pre-monsoon dry sowing. This innovation allows plants under natural farming conditions to extract air moisture for growth. NF farmers practicing APCNF year-round commence with pre-monsoon dry sowing, followed by the main Kharif crop, maintaining relay cropping for a full year. "Pre-monsoon dry sowing" enables cultivation in adverse climates, providing a third crop during summer with minimal water and inputs. Navadanya, a mix of 25–30 crop varieties, comprising millets, oilseeds, pulses, vegetables, tubers, leafy greens, gourds, and flowers, are sown with mulch. In irrigated paddy areas, Navadanya seeds during Kharif reap crop diversity benefits. Extending PMDS, "365 days green cover" enhances livelihoods through diverse food maintenance. Carefully selected crops maximize biodiversity, minimize soil disruption, maintain continuous cover, and sequester carbon through green cover's longevity.

From successful PMDS and green cover piloting, APCNF's A-grade model integrates natural farming principles—365 days of green cover, diverse year-round plants, relay cropping, minimal tillage, mulched residue, seed pelleting, biostimulants, and indigenous seeds. All practices on the same land establish a climate-resilient farming model. 365 days of green cover meant reduced water evaporation losses, continuous living root, diverse microbes in the soil, building of soil organic carbon, and, most importantly, building of water reservoirs in the subsurface soils.

14. Landscape - integration of animals, fisheries, food crops, trees, and horticulture

Integrating animals, fisheries, crops, trees, horticulture, and agroforestry is vital in natural farming. Diverse species, crops, horticulture, trees, livestock, and fisheries are carefully integrated for sustainability. Horticulture trees and field crops together exemplify this. For instance, land divided for horticulture, field crops, and shrubs with varying root depths fulfills nutrient demand and supply. Different species foster diversity, collaboration, and succession. Agroforestry combines tree cover with crop growth, enhancing biodiversity, soil conservation, microclimate regulation, and overall resilience.

15. Cover crops' role on soil health and sustainability

Green cover of the soil contributes directly and indirectly to sustainability and crop productivity as explained below.

Cover crops stabilize soil through diverse mechanisms—boosting infiltration *via* stem and leaf growth—and through root systems. The roots of cover crops, especially legumes, stimulate beneficial fungi that intertwine with soil and release glomalin, binding soil. Some cover crops counteract compaction due to tillage, machinery, and wet soil. Both roots and top growth infuse organic matter into the soil, later decomposing when rolled or tilled. Mulching various crops bolsters soil organic matter in subsurface soils, counteracting its common decline in agriculture. Cover crops are widely endorsed for raising soil organic carbon (SOC) levels, thus improving soil health and climate mitigation. Biomass production of cover crops hinges on species, agronomic management, climate (temperature, precipitation), and soil conditions (moisture, nutrients).

Microbial growth is spurred by added organic matter and growing cover crop roots. The continuous presence of green cover nurtures microorganisms year-round. These organisms suppress diseases, enhance soil structure, and decompose organic matter, freeing nutrients for plants. Diverse crops foster varied microbial populations, enriching soil nutrients. Cover crops that have narrow carbon/nitrogen ratios build bacterial populations, to begin with, followed by fungal populations. The entire soil food web is rebuilt through the promotion of cover crops. According to an internal study conducted by APCNF, the average number of earthworms per square meter in a Natural Farming plot is 46.83 as compared to in the conventional plot where it is 5.71.

Cover crops also balance soil moisture—increasing infiltration, improving structure, and boosting organic matter. They prevent excess nitrogen leaching during heavy rains, acting as "catch" or "trap" crops, releasing nutrients as they decompose for the main crop. Additionally, cover crops fix nitrogen *via* nodules on roots, in cooperation with Rhizobium bacteria.

Incorporating cover crops simplifies pest and disease management. They introduce organic matter, nourishing microbes that aid in disease control. Cover crops attract beneficial insects through their flowers, providing habitat and nectar. In Natural farming, cover crops serve as mulch to suppress weeds. Some covers like buckwheat densely outcompete weeds.

Regarding soil moisture, cover crops are two-edged. In dry climates, they may consume moisture required by the main crop. However, they also stabilize moisture through infiltration, structure, and organic matter. The mulch retains rainwater, even in low-rainfall areas. While benefits surface within the first year, noticeable differences in soil structure and organic matter may take 2–3 years. According to a study conducted by Center for Study of Science, Technology and Policy (CSTEP), switching from conventional to natural farming can potentially save up to an average of 1400–3500 kl of water per ha.

16. Poly cropping practices in natural farming

Since ancient times in India, farmers have embraced practices such as mixed cropping, relay cropping, poly cropping, and multistory farming to diversify crops. In plantation crops, soil vulnerability during early growth stages leads to erosion. Weed infestation poses challenges in these initial stages including fast-growing cover crops that restore soil health.

The APCNF program has, over 7 years, championed mixed crops and cover crops in both irrigation-dependent and rainfed plantation systems, yielding noteworthy results in plant diversity, microbial activity, and soil fertility enhancement. Poly cropping involves cultivating multiple crops on the same plot of land, varying based on agroclimatic zones. Farmers blend crops like vegetables, millets, oilseeds, cereals, climbers, creepers, annuals, and perennials, harmonizing rather than competing. This diversification bolsters resilience against climate-related risks.

17. Soil microbiome

The importance of soil microbiome is supported by several scientific hypotheses. Bringing back microbes into soils is one of the key elements of Natural Farming. Natural Farming practices focus on aboveground and belowground biodiversity in a crop production system throughout the year to maintain the continuous live root system. The decomposition of the above- and below-land plant litter leads to a population of microbes such as fungi, algae, bacteria, earthworms, mites, nematodes, and so forth. Mycorrhizal fungi tubular organisms selectively leach minerals and provide them to plants. One cubic meter of healthy soil has 25,000 Km of fungi. One gram of fertile soil has up to one billion bacteria, and they can swap phosphorus, sulfur, and oxidized iron and carbon. Bacteria use alkaline enzymes and fungi use acidic enzymes to break down organic matter and store nutrients and minerals. Protozoans and Nematodes at an atrophic level in turn consume bacteria and fungi and make nutrients and minerals available in a form that plants can absorb. The microbes get sugars from plants as root exudates, and in turn, microbes leach essential minerals such as manganese, and phosphorus from soils and supply them to plants in inorganic form. All this forms soil organic matter (SOM), which becomes the larger soil food web. Unlike conventional farming, there is no need for supplementing with synthetic fertilizers, which acidify soils and lead to eutrophication. An element like manganese is required for splitting water into H and OH called hydrolysis for photosynthesis, and microbes do that job of making manganese available for plants. While chemically grown food has lost 70% of its nutritional value, the diverse soil biology under Natural Farming is making all 30 to 50 elements available for the plant's growth and sustenance. Soil microbiome and human/ animal gut microbiome are interrelated and connected, and that is where food produced through Natural Farming has all digestible key nutrients. Carbon comes from microbial respiration in the soil where bacteria and other organisms feed on the sugars and release carbon dioxide. Soil organic carbon sequestration is an important outcome of NF practices. SOC sequestration is defined as a state of the soil in which the soil C inputs exceed the soil C release, leading to an increase in SOC stocks. This process helps in fixing atmospheric carbon into soils. Each gram of organic matter when it breaks down due to the above process releases 6 grams of water and is stored in the soil structure. This is a dynamic process; the more the vegetation, the more diverse the microbes, the increased the water availability, and the healthier the plants.

18. Seed microbiome

Seed microbiome offers excellent opportunities for research. Similar to soil microbiome, there are several hypotheses that help in understanding the innovative practice of "seed pelletization" followed in natural farming. Under Natural Farming practices, farmer's own seeds or indigenous are preferred. In conventional farming, it requires at least 17.5 mm of rainfall during 1 week to germinate; additional rainfall is necessary to form seedlings. Additionally, it requires appropriate temperature, moisture, air, and light conditions. Seed pelletization adopted by APCNF farmers treats seeds with beejamrutham (fermented solution of cow dung, cow urine, and lime), a microbial coating, and rolls them with clay and ash to retain moisture for a longer period, and this helps in germination. The pelletization ensures a good microbiome readily available around the seed to immediately get active as soon as the seed becomes physiologically active.

According to Prof James White, Rutgers University, who conducted research on rhizophagy [59] and plant endophytes, the rhizophagy cycle is a process in which microbes switch between an endophytic phase (inside plant roots) and a free-living soil phase. In the rhizophagy cycle, microbes become intracellular and are digested by the plant, releasing nutrients that the plant can use. Endophytes are nonpathogenic microbes that live inside plant tissues without causing harm to the plant. They play a crucial role in the rhizophagy cycle by providing nutrients to the plant and protecting it from pathogens. Plants manipulate bacteria by cultivating microbes on the root exudate zone near the tip of the root. Secretion of exudates in a zone proximal to the root tip facilitates microbe entry into cells of the plant meristem. Inside the plant cells, plants release reactive oxygen (superoxide), which strips the cell walls of the microbes, and nutrients are extracted. Microbes inside the plant trigger a gravitropic response in roots. Microbial protoplasts are expelled into the soil through pores located at the tips of elongating root hairs. These microbes subsequently establish colonies in the soil's rhizosphere, where they obtain supplementary nutrients necessary for their growth and development [59].

19. Traditional crops and varieties

Currently, global crop production is predominantly centered on four staple foods: wheat, rice, maize, and potato. In some regions, this concentration has led to a decline in genetic diversity within crop production and raised concerns about increased susceptibility to diseases. In the past two decades, a staggering 75% of the genetic diversity among agricultural crops has been diminished [60], a 100- to 1000-fold decrease over time. International concerns regarding the depletion of plant diversity have been deliberated in various forums. In 1985, these concerns were addressed by the Commission for Plant Genetic Resources at the Food and Agriculture Organization (FAO). More recently, in 2002, the issue has been revisited during the Conference of the Parties for the Convention on Biological Diversity (CBD), leading to the establishment of GSPC (the Global Strategy for Plant Conservation). The primary goal of GSPC is to rejuvenate plant diversity as an integral part of efforts to eliminate poverty and advance sustainable development. GSPC primarily focuses on preserving seed crops through both *in situ* (in their natural habitat) and *ex situ* (outside their natural habitat) conservation methods.

Traditional varieties are public domain varieties that have a name, developed by ancient farmers, suitable to an ecosystem and farmer needs where they are cultivated and/or conserved by their successors. Farmer needs are diverse and specific to each ecosystem. Resilience to climate and pests, human and domesticated animal nutrition, and taste, cultural, and religious needs determine the selection criteria for these traditional varieties.

Indigenous seeds are better suited to specific regions or circumstances compared to hybrid varieties. Traditional seeds are open-pollinating and generate seeds true to their parent plants. Unlike hybrids, they are robust, resistant to pests, adaptable to their home environment, and require less water and nutrients. Peasant seed multipliers amplify a select few seeds to supply seed banks and fellow farmers. The West Bengal State Seed Corporation Ltd. has distributed 405 MT of desi aromatic rice seeds, marking an extensive institutional distribution of desi rice seeds in India under the Rashtriya Krishi Vikas Yojana (RKVY) scheme. This distribution has covered 20,300 hectares and included eight different varieties of seeds. The purpose of this distribution is to provide enough seeds for seed banks and farmers to cultivate the crops. This is the largest institutional distribution of desi rice seeds in India. In conclusion, traditional agricultural practices offer promising solutions to India's pressing food and sustainability challenges.

Globally, traditional varieties have been viewed by formal plant breeders as source material for plant breeding. Traditional varieties have been exposed over centuries to the vagaries of weather and subjected to a selection process as per the needs of the community beyond food. However, modern agricultural science gave them the status as source material only to develop new and improved varieties. Altieri and Merrick propose that traditional seeds naturally adapt to changing climates through processes like natural crossbreeding and the incorporation of genes from related plants in their local environments, even though they do not provide specific genetic evidence to support this claim. In terms of providing evidence for the resilience capacity of traditional seeds, a genetic-level experiment delved into the gene expressions of these seeds. This experiment centered on three Mexican traditional maize varieties, specifically Michoacan 21, Cajete Criollo, and 85–2, conducted within a greenhouse environment. In this study, scientists meticulously assessed their gene expressions and physiological reactions in their ability to withstand drought stress. Remarkably, the findings unveiled that two out of the three traditional maize varieties demonstrated notable tolerance to drought stress. The performance of traditional varieties in terms of yield and quality is often restricted to the geographies where they have been selected and cultivated.

Mainstreaming of traditional varieties is the need of the hour to infuse resilience to climate and pest complexes and enhance nutrient density and long-term sustainability into agri-food systems. In this direction, pioneering efforts have been made by the Department of Agriculture & Farmer's Empowerment, Government of Odisha by not only coming up with the Standard Operating Procedures to mainstream traditional varieties but also approving the release of four traditional varieties of finger millet that performed better than released variety in terms of grain and straw yield and nutrient density with tolerance to pests and diseases. These varieties are Kundra Bati, Laxmipur Kalia, Malyabanta mami, and Gupteswar Bharati from Koraput, Mangalgiri district of Odisha. Following the example and recognizing the importance, the Department of Agriculture and Family Welfare under the ministry agriculture, Government of India is coming up with a national-level scheme to promote traditional varieties for cultivation through a locally developed seed system coupled with the promotion of natural farming.

20. Research partnerships

Traditional agricultural research has typically operated within rigid disciplinary boundaries, often focusing on narrowly defined goals. However, as there has been a growing emphasis on sustainable agricultural development over the past few decades and a greater recognition of the intricate nature of agricultural and food systems, researchers in this field have sought alternative approaches to effectively leverage scientific progress for broader societal advantages.

Transdisciplinary research emerges as a response to this challenge, as it goes beyond traditional disciplinary confines. This approach transcends and bridges various fields of study, extending even beyond established disciplines. Transdisciplinary research places a strong emphasis on participation and collaboration, facilitating the engagement of natural scientists, social scientists, and diverse stakeholders who may not belong to specific disciplines throughout the entire research process.

When it comes to scaling up practices like Natural Farming in a vast and diverse country like India, successful implementation requires collaborative research projects. These projects aim to generate evidence across various domains, including energy efficiency, water conservation, input reduction, ecosystem services, and the potential for earning carbon credits.

21. NF potential in revamping food systems approach

Food systems around the world have faced significant challenges, particularly highlighted during the COVID-19 crisis when global food supply chains were disrupted due to cross-border movement restrictions. Developing nations, in particular, encounter added difficulties because limited resources and high levels of debt hinder their ability to invest adequately in food systems capable of producing nutritious meals for all segments of society. According to estimates from the United Nations, more than 780 million individuals suffer from hunger, nearly one-third of global food production goes to waste, and nearly 3 billion people cannot afford healthy diets. [61] António Guterres, the United Nations' leader, has urged governments to respond by supporting the UN's call for an SDG Stimulus, which would provide a minimum of $500 billion annually to offer long-term financial assistance to countries in need [62]. Additionally, Mr. Guterres has called for collaboration between governments and businesses, emphasizing the importance of prioritizing people's well-being over profit when building food systems. According to The Global Nutrition Report, 2016, diet is now the number-one risk factor for the global burden of disease [63].

The lessons learned from APCNF present a practical approach to addressing food system challenges. This approach involves strengthening local food supply chains and increasing awareness about the benefits of consuming natural farming (NF) food, which can lead to changes in behavior. While the foundational efforts focus on encouraging farmers to embrace natural farming, the Health and Nutrition (HN) interventions target household consumption habits. A key objective of these interventions is to ensure that the produce from natural farming translates into greater diversity on the dinner plate, enhancing the nutritional value of meals.

One effective strategy is the implementation of low-cost Nutrition Gardens, which serve as scientifically designed kitchen or homestead gardens. These gardens cultivate a variety of nutritious vegetables, fruits, and medicinal plants organically throughout the year, thereby ensuring the nutritional well-being of marginal farming families in rural areas. Women who are part of self-help groups play a pivotal role in promoting natural farming and its benefits to families, particularly targeting groups like adolescent girls, pregnant women, and lactating mothers, thereby influencing the dietary choices of households. The goal is to incorporate 5–7 food groups into the family's diet in all the 129 villages where health and nutrition are being implemented.

APCNF is also actively promoting Nutrition Field Schools as a platform for delivering nutrition education and behavior counseling to communities, with a focus on changing attitudes toward hygiene, health, and nutrition. This initiative aims to include all landless and farmworker groups. Furthermore, APCNF is encouraging the establishment of school kitchen gardens in 178 schools to raise awareness among young children about the health advantages of NF food. The program has facilitated agreements between school management and NF farmers to supply NF food for mid-day meals, aligning with India's commitments made during the UN Food Systems Summit in 2021 under the "School Meals Coalition" alliance. The School Meals Coalition, endorsed by 89 countries, including India, seeks to accelerate efforts to improve and expand school meal programs, ensuring that every child has access to a healthy, nutritious meal at school by 2030 [64]. These initiatives represent a commitment to providing nutritious food and promoting healthier eating habits within communities and schools. According to the True Cost Accounting (TCA) research framework used in the TEEB study, APCNF farms demonstrated 88% higher dietary diversity [64].

Seeding Natural Farming into India's food systems requires systems change to accommodate peer-based learning and extension, revisiting agriculture policies, building institutional capacities, and influencing key stakeholders such as banks and research institutions. APCNF is currently working with the National Council for Educational, Research & Technology to bring changes in the education curriculum by introducing Natural Farming at the school level. The systems change work requires an interdisciplinary approach to mainstream Natural Farming in India's food systems ecosystem. This includes establishing local retail chains, improving community certification processes for residue-free NF food, and revamping the public distribution system.

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